Charge and the electric car

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Explain charge interactions and trace an electric car's energy transfers.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 16.*
It becomes 1/3 as large by Wien's law.

02. Worked example: explain the model
Opposite changes -3 and +3; total charge conserved for the isolated pair.

03. Read and interpret the model
Like signs repel; opposite signs attract. Forces are equal in magnitude and opposite in direction.

04. Apply the physics
Chemical → electrical → rotational → car kinetic energy; thermal energy and sound are possible outputs.

05. Explain your reasoning
The same steady current passes each point; the motor transfers energy, not consuming charge.

06. Review: connect an earlier idea *Review: Unit 6, Day 7.*
+4 mm; partly cancel.

07. Exit check
Repulsion.

08. STRETCH: test the model
No; a charged object can attract a neutral polarizable object. Like net charges repel in the simple model.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 16.*
T≈3000 K (about 3000 K); peak wavelength is rounded.

02. Worked example: explain the model
Opposite changes -3 and +3; total charge conserved for the isolated pair.

03. Read and interpret the model
Like signs repel; opposite signs attract. Forces are equal in magnitude and opposite in direction.

04. Apply the physics
90 J; charge circulates while energy is transferred.

05. Explain your reasoning
The same steady current passes each point; the motor transfers energy, not consuming charge.

06. Review: connect an earlier idea *Review: Unit 6, Day 7.*
Resultant -10 mm. Energy also depends on motion, spatial gradients and the surrounding wave field.

07. Exit check
Polarization redistributes charge; nearer opposite charge can attract more strongly than farther like charge repels.

08. STRETCH: test the model
Shared positive charge gives +4,+4 nC; other remains -8 nC. Total zero.